# Dual Branch Deep Network for Ship Classification of Dual-Polarized SAR Images

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***Abstract*— Ship classification is usually a challenging task due to the small sizes of ship targets and the lack of significant differences between different categories. In terms of synthetic aperture radar (SAR) images, most existing deep learning (DL)-based methods are not designed from the angle of polarimetric characteristics to achieve ship classification. Thus, when facing the ship classification task of dual-polarized SAR images, these networks are often unsatisfactory. To cure this shortcoming, we here propose a novel dual branch deep network (DBDN) specifically designed for dual-polarized SAR ship classification. Our approach consists of three key modules: the image construction module (ICM), the feature extraction module (FEM), and the feature fusion and classifier module (FFCM). In ICM, two novel pseudo-RGB images are constructed for the first time, i.e., the polarimetric features-guided pseudo-RGB image (PF-RGB) and the texture features-guided pseudo-RGB image (TF-RGB), which can more accurately and comprehensively reflect ships' characteristics. FEM enables the network to focus on important ship features and suppress irrelevant noise through transferred layers and designed ConvNeXt-Attention block (CNABlock), enhancing the discriminative capability of different ships. Finally, FFCM extracts and combines various ship features for classification, wherein the enhanced inverted residual block (EIRBlock) and the channel spatial attention module (CSAM) components are proposed as well. The performance of DBDN is evaluated on the OpenSARShip2.0 dataset, and experimental results show that DBDN achieves excellent performance in all evaluation metrics in comparison with some state-of-the-art (SOTA) algorithms. For example, compared to the recently proposed method deep SAR-net (DSN), DBDN further improves the accuracy by 4.74% and 4.07% in the three-class and six-class classification tasks, respectively.**

***Index Terms*— Dual branch deep network (DBDN), dual-polarized synthetic aperture radar (SAR), feature extraction module (FEM), feature fusion and classifier module (FFCM), image construction module (ICM), ship classification.**

## I. Introduction

SYNTHETIC aperture radar (SAR) is an advanced radar system that utilizes radar beam synthesis techniques to obtain high-resolution images regardless of the weather and illumination, which has become a crucial tool for Earth observation [1]. In particular, rapid detection and classification of ship targets using SAR images are necessary and meaningful for maritime activities, such as sea rescue and transportation management [2], [3].

Briefly speaking, the objective of SAR ship classification is to categorize and identify ships using their intrinsic characteristics, including backscattering intensity, reflection properties, shape, and size. Keeping this in mind, various methods have been developed. Among them, deep learning (DL) enables networks to adaptively learn more robust and effective features, making it the mainstream way to realize ship classification [4]. In the early study, Bentes et al. [5] only designed a simple convolutional neural network (CNN) to classify ships, such as cargo and tanker. Then, with the development of CNN and DL, more and more complicated networks are constructed. For example, He et al. [6] improved a triple network to learn deeper features, wherein triple loss terms were used to increase interclass distance. Wang et al. [7] utilized the multiscale features and attention mechanisms to suppress clutter and classify ships. To achieve fast and efficient ship classification, Zhang et al. [8] incorporated the squeeze-and-excitation (SE) module into ShuffleNet V2, and Jiang et al. [9] combined the spiking neural network with the Siamese network. To address the challenge of limited labeled samples in SAR images, Xu and Lang [10] proposed a method known as geometric transfer metric learning, which introduced transfer metric learning into SAR ship fine-grained classification for the first time. Zheng et al. [11] proposed a novel heterogeneous network named MetaBoost, which can automatically combine different classifiers to achieve the optimal classification results.

To further enhance the classification accuracy, in recent works, researchers also introduced some additional information into the ship classification task. For instance, traditional manually extracted features of single polarization SAR images, like local radar cross section (LRCS) [12] and histogram of oriented gradient (HOG) [13], were reasonably exploited to improve the ship classification performance of networks. Xie et al. [14] proposed the HOG-SiamShipCLSNet, which integrated features extracted by HOG operator with features extracted using CNNs. Different from them, some work directly considered the dual-polarized SAR images as the inputs to guide the construction of networks. Specifically, Xiong et al. [15] first used a lightweight Hourglass network to prescreen ship targets and then employed the EfficientNet-B2 backbone with shared weights. Zeng et al. [16] proposed a

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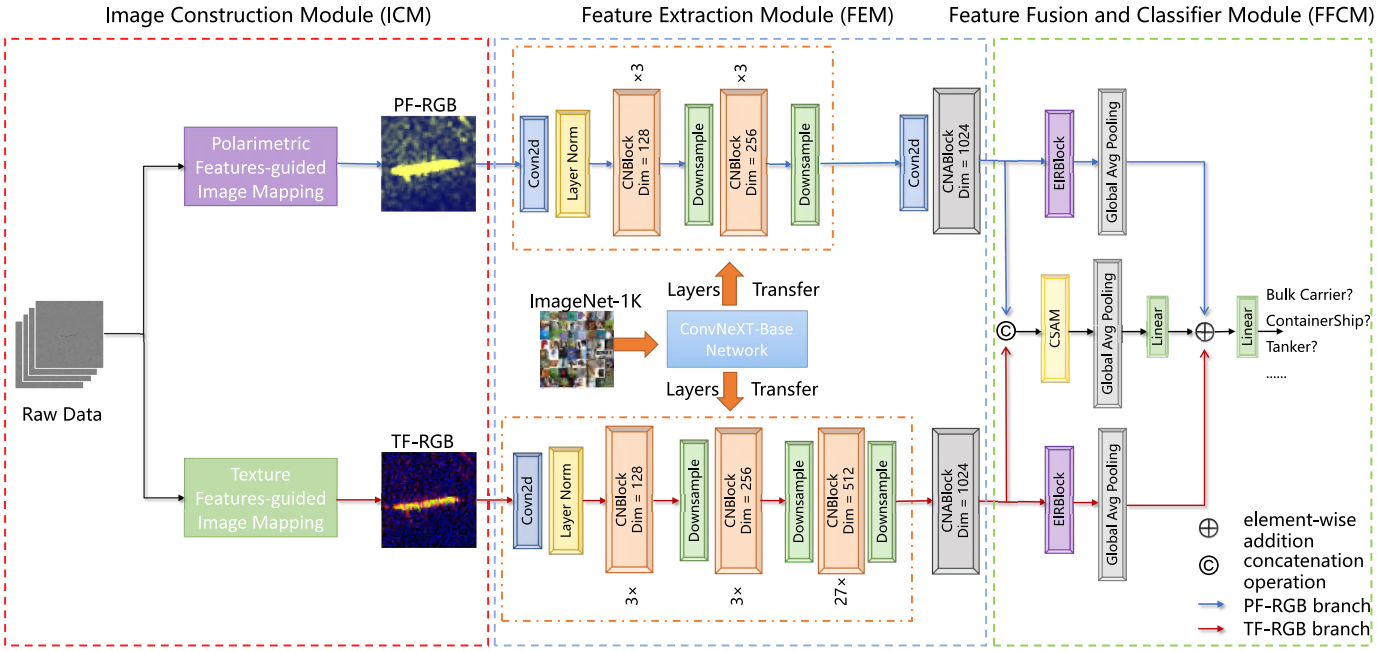



Fig. 1. Framework of DBDN.

hybrid channel feature loss to classify ships. He et al. [17] put forward the bilinear CNN (BCNN) for dual-polarized ship classification. Although these methods have shown the effectiveness in ship classification, they only paid attention to the utilization of amplitude features.

Overall, aforementioned methods are more concerned on the designs of network structures, neglecting the digging of the inherent polarimetric characteristics of SAR images. In fact, the polarimetric information is specific for polarimetric images. The reasonable use of the polarimetric features extracted from these information is also beneficial for ship classification. Motivated by this, Zhang [18], Zhang [19], and Shao et al. [20] incorporated the polarization coherence feature into the networks for further improving their ship classification performance. Via performing joint time–frequency analysis (JTFA) on the original complex SAR images, Huang et al. [21] used a set of radar spectrogram for more completely reflecting the polarimetric characteristics of ships. Even so, existing works related to the introduction of polarimetric features into networks for ship classification are still few. In other words, how to effectively classify SAR ship targets using polarimetric features-related networks is still an open issue deserving research. Datcu et al. [22] highlighted how integrating the polarized characteristics into DL frameworks can significantly improve the interpretability and reliability of such models, thereby advancing their applicability in target recognition tasks. Radman et al. [23] proposed dual-branch fusion of miniGCN and CNN networks (FuNet) for PolSAR image classification, which feeds spatial and polarimetric features into two branches to fully exploit the rich information of PolSAR images.

Inspired by above methods, we here propose a dual branch deep network (DBDN) for dual-polarized SAR ship classification, which includes three modules, i.e., image construction module (ICM), feature extraction module (FEM), and feature fusion and classifier module (FFCM). Considering that different features can depict ships' different characteristics, ICM is mainly designed to extract and integrate polarimetric and texture features as input data. FEM is constructed to extract deeper features with the transfer technology and attention mechanism. The last module FFCM is proposed to further extract and fuse the features obtained from dual branches for classification. The overall architecture is shown in Fig. 1.

Overall, the contributions of this article are threefold.

1) An ICM is designed to more precisely and completely reflect ships' characteristics, wherein two novel pseudo-RGB images, i.e., the polarimetric features-guided pseudo-RGB image (PF-RGB) and the texture features-guided pseudo-RGB image (TF-RGB), are constructed for the first time.
2) To further dig and utilize the abstract information of dual-polarized SAR images, the other two modules named FEM and FFCM are proposed. The former focuses the enhancement of attention information, yet the latter concerns on the utilization of attention information jointly with other information.
3) One new network DBDN is developed based on ICM, FEM, and FFCM for ship classification. Also, the superiority of DBDN over the other state-of-the-art (SOTA) networks is confirmed by the experiments performed on the OpenSARShip2.0 dataset.

The remaining of this article is organized as follows. Section II is the theoretical background. Section III provides a detailed introduction of proposed method. The experiments and analyses are presented in Section IV. Section V concludes this article.

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## II. Theoretical Background

In this section, the degree of polarization (DoP), the polarimetric entropy, and the grayscale and rotation invariant local binary pattern (LBP) operator are, respectively, introduced.

### A. Degree of Polarization

For a given incident polarization, the Stokes parameters of backscattered field are usually obtained by [24]

$$\begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} |E_H|^2 + |E_V|^2 \\ |E_H|^2 - |E_V|^2 \\ 2\,\mathrm{Re}\left(E_H E_V^*\right) \\ -2\,\mathrm{Im}\left(E_H E_V^*\right) \end{bmatrix} \tag{1}$$

where $\{g_0, g_1, g_2, g_3\}$ are the Stokes parameters, $E$ is the (complex) voltage, the subscripts $H$ and $V$, respectively, represent the horizontal polarization and vertical polarization, $|\cdot|$ represents the absolute value, $\mathrm{Re}(\cdot)$ is the real part, $\mathrm{Im}(\cdot)$ is the imaginary part, and $^*$ represents the complex conjugate. Then, with $\{g_0, g_1, g_2, g_3\}$, the DoP of electromagnetic wave is expressed as [25]

$$\mathrm{DoP} = \frac{\sqrt{\langle g_1\rangle^2 + \langle g_2\rangle^2 + \langle g_3\rangle^2}}{\langle g_0\rangle} \tag{2}$$

where $\langle\cdot\rangle$ stands for the temporal averaging. The range of DoP is [0, 1].

### B. Polarimetric Entropy

The Jones coherency matrix $\boldsymbol{J}$ is defined as [26]

$$\begin{aligned} \boldsymbol{J} &= \begin{bmatrix} \langle E_H E_H^*\rangle & \langle E_H E_V^*\rangle \\ \langle E_V E_H^*\rangle & \langle E_V E_V^*\rangle \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} \langle g_0\rangle + \langle g_1\rangle & \langle g_2\rangle - j\langle g_3\rangle \\ \langle g_2\rangle + j\langle g_3\rangle & \langle g_0\rangle - \langle g_1\rangle \end{bmatrix}. \end{aligned} \tag{3}$$

Via computing the eigenvalues and eigenvectors of $\boldsymbol{J}$, we can obtain its diagonalized form. Physically, the diagonalized $\boldsymbol{J}$ matrix reflects the statistical independence between a pair of electromagnetic wave components, which can be expressed as [27]

$$\boldsymbol{J} = \boldsymbol{U}_2 \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \boldsymbol{U}_2^{-1} = \lambda_1 \underline{u}_1 \underline{u}_1^{T^*} + \lambda_2 \underline{u}_2 \underline{u}_2^{T^*} \tag{4}$$

where $\boldsymbol{U}_2 = [\underline{u}_1, \underline{u}_2]$ represents a $2 \times 2$ unitary matrix belonging to the SU(2) group [28]. It consists of a pair of orthonormal eigenvectors. The eigenvalues, $\lambda_1 \geqslant \lambda_2 \geqslant 0$, are nonnegative real values and can be calculated using the following equation [29]:

$$\begin{aligned} \lambda_1 &= \frac{1}{2}\left\{\langle g_0\rangle + \sqrt{\langle g_1\rangle^2 + \langle g_2\rangle^2 + \langle g_3\rangle^2}\right\} \\ \lambda_2 &= \frac{1}{2}\left\{\langle g_0\rangle - \sqrt{\langle g_1\rangle^2 + \langle g_2\rangle^2 + \langle g_3\rangle^2}\right\}. \end{aligned} \tag{5}$$

Then, the polarimetric entropy ($H$) is defined as [27]

$$H = -\sum_{i=1}^{2} p_i \log_2 p_i, \quad \text{where } p_i = \frac{\lambda_i}{\lambda_1 + \lambda_2}. \tag{6}$$

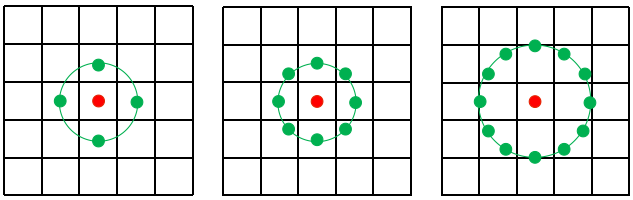


Fig. 2. Circularly symmetric neighbor sets for different $(P, R)$ values. (a) $P = 4$ and $R = 1.0$. (b) $P = 8$ and $R = 1.0$. (c) $P = 12$ and $R = 1.5$.

### C. Gray-Scale and Rotation Invariant LBP

The original LBP operator mainly compares the gray value differences between the central pixel and its peripheral eight pixels within a $3 \times 3$ region [30]. In order to adapt to texture features of different scales and meet the requirements of gray-scale and rotation invariance, Ojala et al. [31] further improved the LBP operator by extending the $3 \times 3$ neighborhood to any neighborhood, i.e., replacing the square region with a circular region, which was named $\mathrm{LBP}_{P,R}$. Its circular region is illustrated in Fig. 2, and calculation process is given below

$$\mathrm{LBP}_{P,R} = \sum_{p=0}^{P-1} s\left(g_p - g_c\right)2^p, \quad s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{7}$$

wherein the number of pixels in the local neighborhood is $P(P > 1)$, $g_c$ represents the gray-scale value of central pixel, $g_p$ $(p = 0, \ldots, P-1)$ represents the gray-scale value of the pixel on the circle with a radius of $R(R > 0)$ around the central pixel, and $s(\cdot)$ is the sign function.

To achieve rotation invariance, $\mathrm{LBP}_{P,R}$ is then modified by selecting the minimum value among all possible rotations, which ensures that the resulting pattern remains the same regardless of the rotation angle, that is

$$\mathrm{LBP}_{P,R}^{\mathrm{ri}} = \min\left\{\mathrm{ROR}\left(\mathrm{LBP}_{P,R}, i\right) | i = 0, 1, \ldots, \mathrm{P}-1\right\} \tag{8}$$

where $\mathrm{ROR}(\mathrm{LBP}_{P,R}, i)$ is a rotation function, indicating that $\mathrm{LBP}_{P,R}$ is rotated to the right by $i$ bits. Via selecting the minimum value among all possible rotations, a unique identifier can be assigned to each rotation-invariant LBP, thereby achieving rotation invariance. For more details, we refer readers to [31].

## III. Methodology

In this section, we will introduce the novel dual-polarized SAR ship classification method DBDN, which includes three modules, i.e., the ICM, the FEM, and the FFCM.

### A. Image Construction Module

The ICM is mainly designed to obtain two different input maps, i.e., the PF-RGB and the TF-RGB. They are explained in detail in the following.

*1) PF-RGB:* Polarimetric information has been widely applied to ship detection [2], [32], [33]. But, for the ship classification task, only a few works have been done so far with this kind of information, wherein the mainstream way is the separate or joint utilization of the amplitudes of VV and

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VH channels [15], [16], [34]. Actually, the classification performance of these amplitude-based methods is limited, since the polarimetric differences between different kinds of ships cannot be reflected by amplitude, especially the information related to phase, like the co-polarized phase difference. So, this implies that the polarimetric features should be considered if we want to attain more satisfactory ship classification results. To this end, we here propose the PF-RGB image, whose combination manner is

$$\begin{aligned}\text{PF-RGB}(m,n,1) &= H(m,n)\\ \text{PF-RGB}(m,n,2) &= \text{DoD}(m,n)\\ \text{PF-RGB}(m,n,3) &= \text{RS}(m,n)\end{aligned} \tag{9}$$

wherein PF-RGB $(m, n, i)$ $(i = 1, 2, 3)$ means that the size of the $i$th channel of PF-RGB is $m \times n$. DoD and RS, respectively, denote the degree of depolarization and the reflection symmetry, and their expressions are

$$\begin{cases}\text{DoD} = 1 - \text{DoP}\\ \text{RS} = |\langle E_V E_H^* \rangle|.\end{cases} \tag{10}$$

The reason and effectiveness behind this combination are given below.

1) *H:* The polarimetric entropy $H$ is usually exploited to represent the degree of backscattering uncertainty and randomness [33]. As one kind of man-made target, ships often have complex metal structures, such as the hull, superstructure, and masts [35]. So, these different structures can cause different backscatterings. Meanwhile, existing ship types are also various, each with its own unique structural components. Therefore, analyzing the variability and randomness of backscattering mechanisms can allow us to distinguish different ship types. As a result, $H$ is apt for ship classification.
2) *DoD:* Within existing polarimetric theories, DoP is one of the most important parameters related to the partial polarization state of scattered wave [25]. Conversely, DoD corresponds to the DoP of scattered waves, which has not been widely utilized for ship classification. It is known that different materials have different dielectric constants and electromagnetic characteristics, which can affect the propagation and interaction of electromagnetic waves [29]. In terms of ships, their structural materials could be different from each other. So, this difference can easily make ships' reflected electromagnetic waves different, resulting in different DoD values. Thus, using DoD to classify ships is feasible.
3) *RS:* As one important polarimetric feature, RS is widely adopted to distinguish ships from sea clutter. In fact, RS is one structure-related feature as well, because it describes the correlation between like- and cross-polarization channels that often exists in the reflection asymmetry man-made targets [36]. So, from the viewpoint of ship classification, it can highlight different structure information of different ships. On the other hand, RS has been demonstrated to have a certain level of robustness to environmental variables, such as incident angle and sea conditions [36]. So, it has a natural advantage over ship classification. As a consequence, we choose it as the last parameter to construct the PF-RGB image.

Obviously, via fusing these three polarimetric features together, we can more effectively utilize the polarimetric difference characteristics between ships for classification. To make readers better understand the reasonability, we here also choose three different types of ships (i.e., bulk carrier, container ship, and tanker) for visual explanations (see Fig. 3). It is evident that the chosen three polarimetric features can really better highlight the differences in the details of ship's structures than the VV and VH amplitude maps. Thus, it is reasonable for us to adopt them to form the PF-RGB image for ship classification.

*2) TF-RGB:* Combining traditional manual features with CNNs is recently popular for SAR image classification [12], [13] as we said before. This is because compared to the high-level features obtained from networks, traditional low-level image features have the advantage of capturing fine-grained details and local patterns. Particularly, Zhao et al. [37] investigated the performance of six traditional features, among which the LBP operator achieved the best comprehensive classification results. For hyperspectral image classification, Zhang et al. [38] also achieved a satisfactory result based on LBP.

In essence, the LBP operator captures local texture patterns by comparing the intensity values of pixels in a neighborhood. Among its variants, the gray-scale and rotation invariant $\text{LBP}_{P,R}$ introduced in Section II is widely used, allowing for the robust representation of targets in various orientations and intensities. Furthermore, the LBP operator also has a superior ability to extract features while maintaining spatial consistency with the original image, which can facilitate the direct integration of texture features at the input stage, obviating the need for complicated fusion strategies. Motivated by these advantage, hereafter, we introduce the $\text{LBP}_{P,R}$ operator into the ship classification task.

In brief, we combine $\text{LBP}_{P,R}$ with the VV and VH amplitude maps to create the pseudotexture RGB image, whose construction process is similar to that of PF-RGB, that is

$$\begin{aligned}\text{TF-RGB}(m,n,1) &= \text{VV}(m,n)\\ \text{TF-RGB}(m,n,2) &= \text{VH}(m,n)\\ \text{TF-RGB}(m,n,3) &= \text{LBP}_{P,R}(m,n).\end{aligned} \tag{11}$$

Here, $\text{LBP}_{P,R}$ is extracted from the VV amplitude map. The rationality behind this combination is that the channels VV and VH have been extensively demonstrated to be effective in ship classification [15], [16], [34]. Meanwhile, handcrafted features are often designed based on domain expertise and experience, which can capture specific discriminative patterns or characteristics that may not be effectively learned from amplitude maps by deep neural networks. As shown in Fig. 3(g)–(i), the features extracted by $\text{LBP}_{P,R}$ can indeed highlight the contours of ships that are different from the VV amplitude map. So, the combination of $\text{LBP}_{P,R}$ with the VV and VH amplitude maps enables us to take advantage of the strengths of both and

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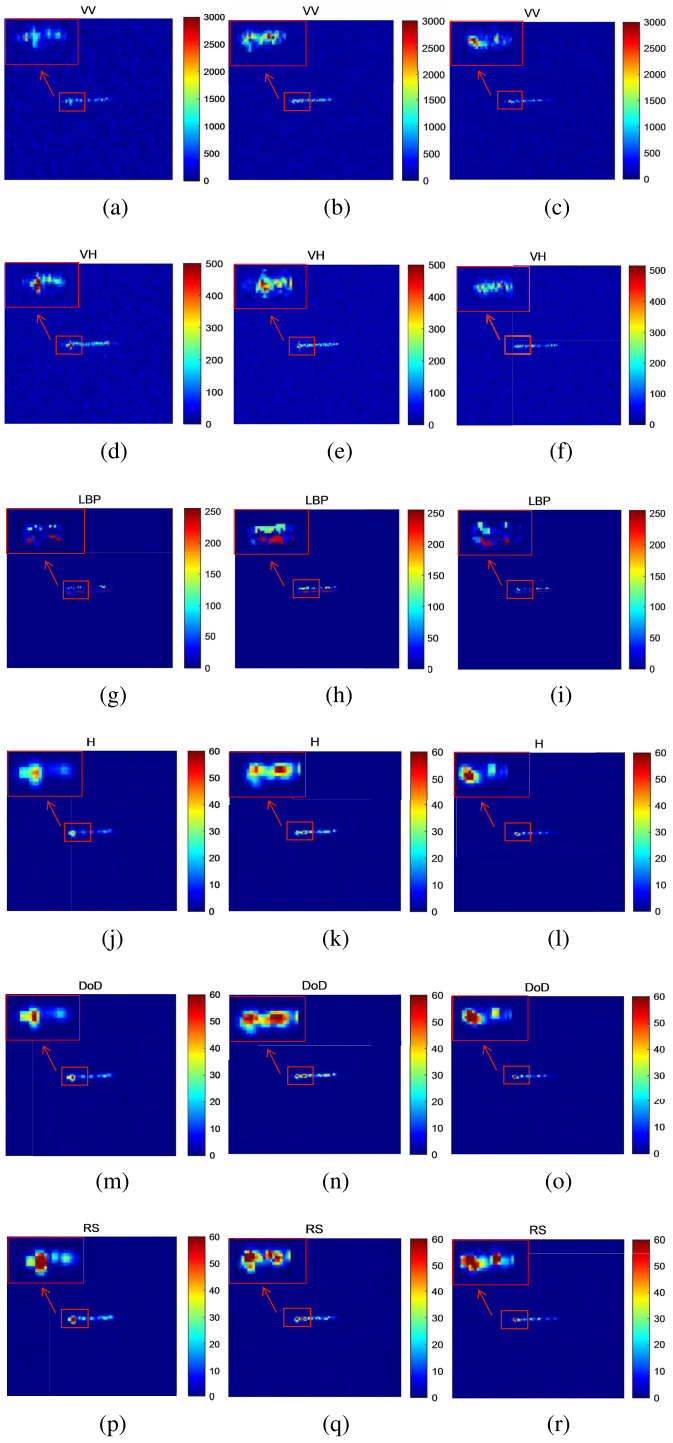



Fig. 3. VV/VH amplitude maps, $\mathrm{LBP}_{P,R}$ features, and polarimetric feature maps. (Left column) Bulk carrier. (Middle column) Container ship. (Right column) Tanker. (a)–(c) VV amplitude maps. (d)–(f) VH amplitude maps. (g)–(i) $\mathrm{LBP}_{P,R}$. (j)–(l) $H$. (m)–(o) DoD. (p)–(r) RS.

obtain a more comprehensive and robust feature representation for ship classification.

Overall, the structures of pseudo-RGB images are the same as that of true RGB images, namely, both have three channels. Therefore, the existing true RGB image-based classification methods and techniques can be applied to our proposed pseudo-RGB images. On the other hand, via representing polarimetric or texture features in the form of RGB image,

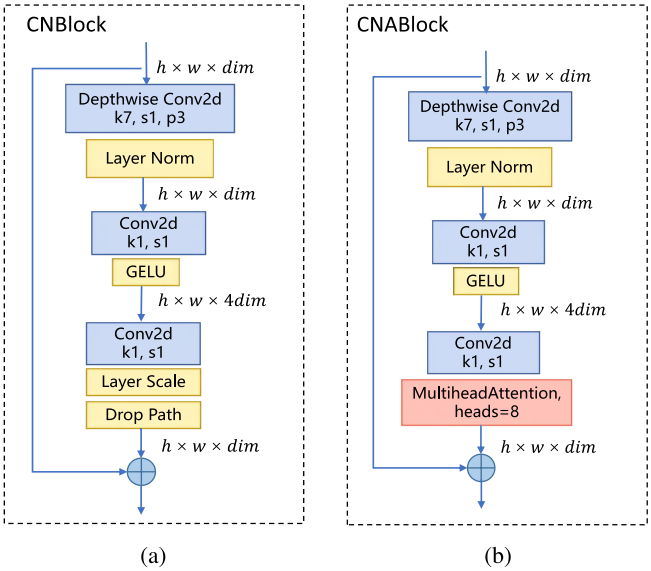



Fig. 4. Network blocks in FEM. (a) CNBlock. (b) CNABlock.

the spatial relationships between these features are effectively captured. This comprehensive representation facilitates more thorough analyses of the structural and polarimetric characteristics of targets through channel correlation analyses. In turn, the incorporation of such spatial relationships enhances the discriminative capability of the classification model, which will be validated in the later experiments. As a result, our proposed module ICM is helpful for ship classification.

### B. Feature Extraction Module

When facing a limited number of training samples, a cost-effective solution is to perform fine-tuning with a pretrained model from other large-scale labeled dataset, i.e., transfer learning. In [39], a detailed investigation was carried out on the application of transfer learning to various SAR data-based tasks, from which the conclusion that low-level features extracted from large-scale natural images can effectively represent SAR targets was obtained. Furthermore, it was observed that as the depth and width of network increase, the learned low-level features become more universally applicable. Given these, to alleviate the issue of small samples in ship classification, we introduce the transfer learning technology into our method as well. Specifically, the network ConvNeXt [40] trained on ImageNet is here chosen as the backbone of our FEM, wherein the CNBlock composed of an inverted bottleneck is the core [see Fig. 4(a)].

Assuming that the input feature is $x$, the output $y$ obtained through CNBlock can be represented as

$$y = \mathscr{D}\{\mathscr{L} * \mathcal{F}(x)\} + x \tag{12}$$

where $\mathcal{F}(\cdot)$ represents the overall mapping of the inverted bottleneck, $\mathscr{L}$ denotes the scaling factor, and $\mathscr{D}$ controls the probability of retaining each model layer.

However, CNBlock is designed for natural image classification, relies on the convolutional and linear layers, and has obvious limited capability in handling complex background

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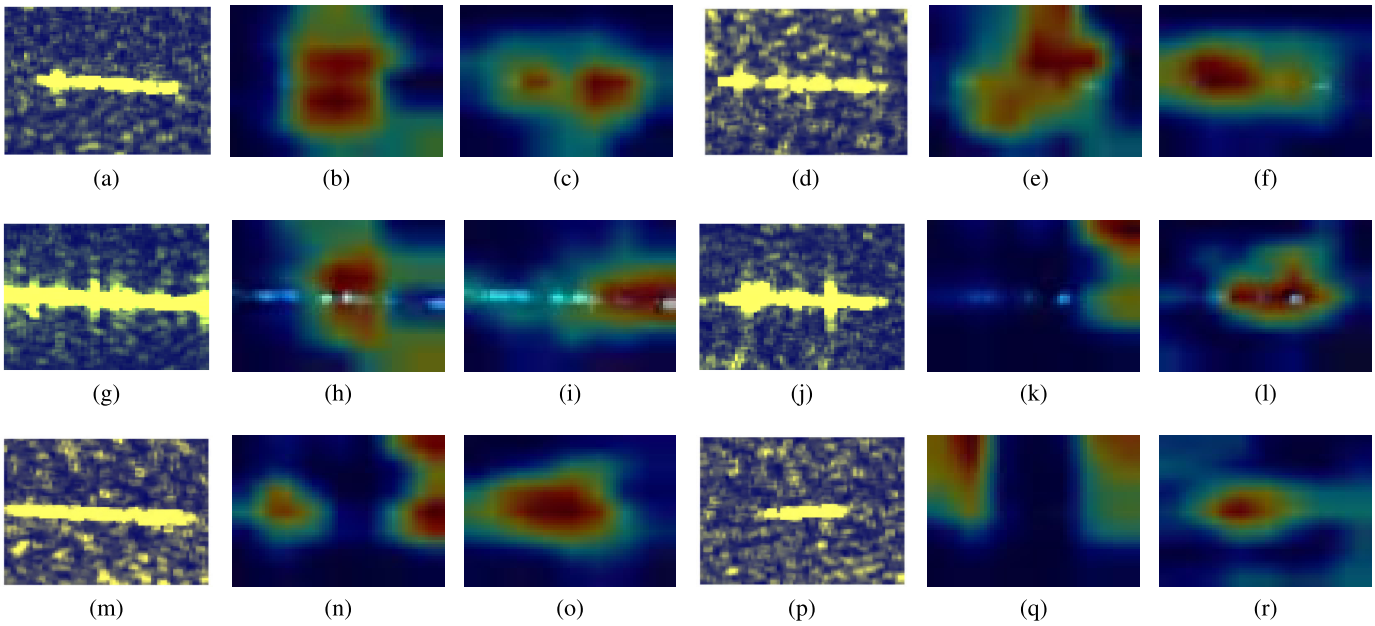



Fig. 5. PF-RGB and CAM images of different types of ships. The first row is bulk carrier [(a)–(f)], the second row is container ship [(g)–(l)], and the third row is tanker [(m)–(r)]. Specifically, the first and fourth columns are the PF-RGB images of ships [i.e. (a), (d), (g), (j), (m) and (p)]; the second and fifth columns represent the CAM output features of CNBlock [i.e. (b), (e), (h), (k), (n), and(q)], while the third and sixth columns represent the CAM output features of the same model after replacing the last CNBlock with CNABlock [i.e. (c), (f), (i), (l), (o) and (r)].

interference and capturing intricate relationships between features of different scales. For SAR images, they are often characterized by a significant amount of clutter and background interference. In terms of SAR ship classification task, ships usually exhibit variations in shape and appearance depending on viewpoints, and the ships of same type could also have different scales (see Fig. 5). Fortunately, attention mechanisms have been proven to be highly effective in SAR ship classification [7], [41], which enables the module to focus on relevant regions of the image and adaptively focus on features at different scales. So, to further improve the performance of CNBlock, we here propose a ConvNeXt-Attention block (CNABlock) module via introducing the self-attention mechanism, which can be mathematically expressed as

$$y = \mathscr{A}\{\mathcal{F}(x)\} + x \tag{13}$$

where $\mathscr{A}(\cdot)$ is the multihead attention, which can be expressed as [42]

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{14}$$

where $\text{head}_i$ $(i = 1, \ldots, h)$ refers to a specific instance that executes a distinct attention calculation, concentrating on diverse aspects of the input data. $Q$, $K$, and $V$ represent the queries, keys, and values, respectively. The output weights are determined by the parameter matrix $W^O$. In this work, we employ $h = 8$. More details of multihead attention mechanism can be found in [42]. Note that the scaling factor and dropout regularization technique are not used in CNABlock. By integrating the self-attention mechanism into CNBlock, the network is able to focus on significant ship features and inhibit unnecessary background interference. This consequently improves its capacity to acquire intricate details and complex correlations between features.

Without loss of the generality, also, we present the class activation map (CAM) [43] in Fig. 5 to further illustrate the effectiveness of CNABlock. The CAM can effectively reflect the importance of features in the network's decision-making process by weighting and summing the feature maps from a particular network layer. For example, a bulk carrier is shown in Fig. 5(a) using PF-RGB, and Fig. 5(b) and (c) shows its CAMs obtained from CNBlock and CNABlock, respectively. Obviously, the thermal region in Fig. 5(b) is mainly distributed around the center area of ship, whereas in Fig. 5(c), the distribution along the whole ship structure is slender. From Fig. 5(a), it is also seen that the bulk carrier is slender. This means that the CAM obtained from the model incorporating CNABlock shows a more focused and relevant activation in the target area, contributing to the enhanced classification performance. The same conclusion can be obtained by analyzing Fig. 5(d)–(r).

Once CNABlock is obtained, the module FEM in Fig. 1 will be constructed with two different branches, i.e., the PF-RGB branch (i.e., blue arrows) and the TF-RGB branch (red arrows). For both, the manner of layer organization used in ConvNeXT-Base is introduced, viz., the orange dash–dotted boxes. However, our method differs from ConvNeXT-Base in that we reduce the number of layers, resulting in the migration of only low-level features from natural images, as previously explained. After obtaining universal features, the data are further processed by CNABlock to accurately extract ship targets from the clutter interference in the SAR image. To achieve dimensional alignment, a convolutional layer is added to the PF-RGB branch. By doing so, both PF-RGB and TF-RGB branches of FEM can effectively utilize the generalization information of natural images and capture the unique features of each input.

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### C. Feature Fusion and Classifier Module

This module is mainly designed to further extract and fuse the features for classification, wherein two feature extraction components [i.e., enhanced inverted residual block (EIRBlock) and channel spatial attention module (CSAM)] and one feature fusion strategy are jointly adopted. They are explained in detail in the following.

*1) EIRBlock:* Reanalyzing Fig. 5, one can find that the region of interest is more concentrated after the utilization of CNABlock. So, we can deem it as the input to the following module FFCM, which also need to be further explored for deeper feature extraction. As component of the lightweight network MobileNetV2 [44], IRBlock incorporates the innovative concept of "inverted residuals and linear bottlenecks." It can effectively reduce computational complexity and enhance feature expression, which is suitable for various computer vision tasks. However, when applying the original IRBlock to SAR ship classification, there exist some issues, for example, as follows.

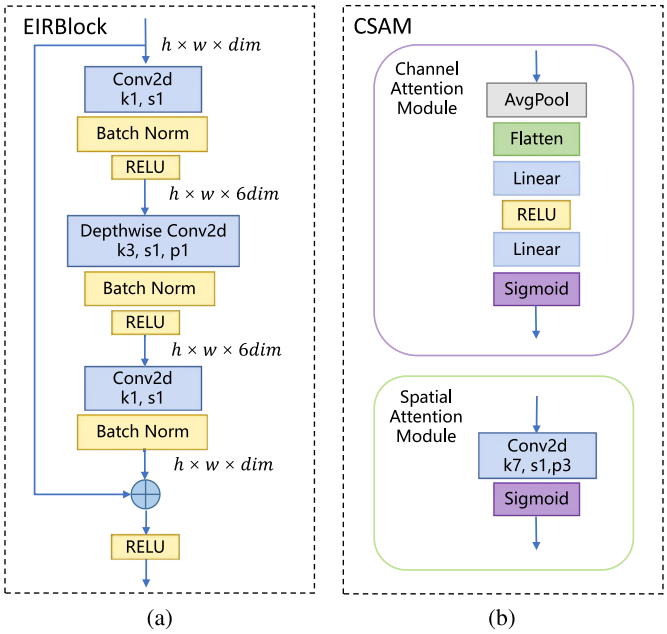
1) In the context of SAR ship classification, preserving fine-grained feature details is crucial for accurate identification. Unfortunately, IRBlock utilizes rectified linear unit 6 (ReLU6) as the activation function, which truncates both negative (<0) and positive (>6) values. Without doubt, this can also lead to the loss of important polarimetric feature information.
2) The absence of an activation function after the residual connections can incur the increasing computational overhead.

To cure these shortcomings, we first replace the ReLU6 activation function with the standard ReLU in IRBlock. By allowing for a broader positive range, ReLU preserves more fine-grained details, enabling the network to capture subtle variations in SAR ship images and improve classification performance. Then, we also add a ReLU activation function after the residual connections in IRBlock. This adjustment not only reduces the computational complexity but also enhances the network's ability to learn discriminative features. Finally, EIRBlock is illustrated in Fig. 6(a), which can be mathematically expressed as

$$\mathcal{F}_I(x) = [A \circ \mathcal{N} \circ B]x \tag{15}$$

where $x$ is the input feature map. $A$ is a linear transformation that transforms $x$ from $\mathcal{R}^{s\times s\times k}$ to $\mathcal{R}^{s\times s\times n}$, where $k$ and $n$ both represent the number of feature channels. $k = t \times n$, and $t$ is the expansion factor. $\mathcal{N}$ is a nonlinear channel by channel transformation that transforms the input from $\mathcal{R}^{s\times s\times n}$ to $\mathcal{R}^{s'\times s'\times n}$, where the transformation from $s$ to $s'$ is determined by the stride of convolutional layer. Here, $\mathcal{N} = \text{ReLU} \circ \text{dwise} \circ \text{ReLU}$. $B$ is another nonlinear transformation that transforms the input from $\mathcal{R}^{s'\times s'\times n}$ to $\mathcal{R}^{s'\times s'\times k'}$, where $k'$ represents the number of output channels. Through the combination of linear and nonlinear transformations, EIRBlock can extract more discriminative features.

*2) CSAM:* EIRBlock extracts features from each branch separately, without considering the potential associations between different features. To address this limitation, we draw inspiration from the SE [45] and CBAM [46] modules, and hereafter design another feature extraction component CSAM, which aims to extract joint features that are concatenated on the channel dimension. The specific structure of CSAM is shown in Fig. 6(b).




(a) (b)

Fig. 6. Network blocks in FFCM. (a) EIRBlock. (b) CSAM.

Given an intermediate feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ as input, CSAM sequentially calculates a 1-D channel attention map $\mathbf{M_c} \in \mathbb{R}^{C\times 1\times 1}$ and a 2-D spatial attention map $\mathbf{M_s} \in \mathbb{R}^{1\times H\times W}$. The overall attention process can be summarized as

$$\begin{aligned} \mathbf{F}' &= \mathbf{M_c}(\mathbf{F}) \otimes \mathbf{F} \\ \mathbf{F}'' &= \mathbf{M_s}(\mathbf{F}') \otimes \mathbf{F}' \end{aligned} \tag{16}$$

where $\otimes$ denotes elementwise multiplication and $\mathbf{F}''$ is the final refined output. The channel attention map $\mathbf{M_c}$ can be represented using the following mathematical formula:

$$\mathbf{M_c} = \sigma(\mathcal{L}(\text{ReLU}(\mathcal{L}(\text{Flatten}(\mathcal{AP}(\mathbf{F})))))) \tag{17}$$

where $\sigma(\cdot)$ represents the Sigmoid function, $\mathcal{L}(\cdot)$ represents the fully connected layer, Flatten represents the flatten operation, and $\mathcal{AP}(\cdot)$ represents the adaptive average pooling operation. $\mathbf{M_s}$ is the spatial attention map that is represented with the formula

$$\mathbf{M_s} = \sigma\left(\mathbf{Conv2d}(\mathbf{F}')\right) \tag{18}$$

where $\mathbf{Conv2d}(\cdot)$ represents the convolutional layer.

The rationality behind CSAM is explained below. SAR ship images often exhibit complex structures and varying sizes, containing rich detailed information. The channel attention map $\mathbf{M_c}$ of CSAM mainly focuses on the relationships between different channels. By learning the weights of each channel, it enhances the useful channels while suppressing the irrelevant ones, thereby improving the channel characteristics of ships. On the other hand, the spatial attention map $\mathbf{M_s}$ of CSAM learns the importance of different positions within

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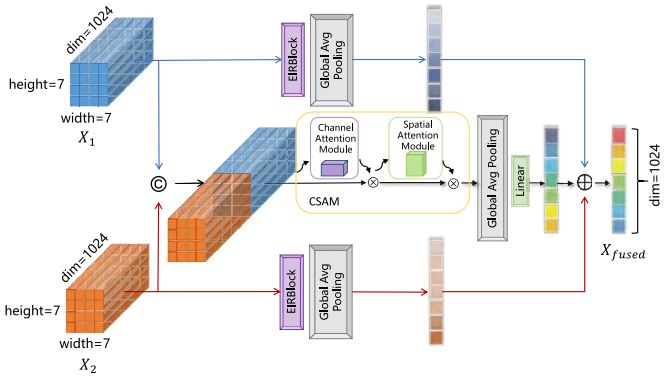



Fig. 7. Schematic of feature fusion module.

the feature map so as to enhance the spatial contextual information. Thus, these two attention mechanisms working together can improve network's ability to accurately locate and distinguish targets, enhancing the ship classification accuracy.

*3) Feature Fusion Strategy:* The proposed fusion strategy is mainly designed to fuse the individual features and combined features together, as shown in Fig. 7. Here, the individual features are the features separately processed by EIRBlock, and the combined features are the features processed by CSAM. The whole feature fusion strategy is mathematically expressed as

$$X_{\text{fused}} = \mathcal{AP}(\mathcal{F}_I(X_1)) + \mathcal{AP}(\mathcal{F}_I(X_2)) + \mathcal{L}(\mathcal{AP}(\mathcal{F}_C(X_1 \copyright X_2))) \quad (19)$$

where $X_1$ represents the PF-RGB branch output features and $X_2$ represents the TF-RGB branch output features. $\mathcal{F}_I$ is the mapping of EIRBlock, $\mathcal{F}_C$ represents the mapping of CSAM, and © denotes the concatenation operation. The final fused representation $X_{\text{fused}}$ is obtained by elementwise adding the outputs.

Finally, the fused features are fed into the classifier for classification.

## IV. Experiments and Discussion

In this section, the performance of DBDN will be evaluated. For this goal, three-classification and six-classification experiments are, respectively, performed on DBDN using the OpenSARShip2.0 dataset. Besides, ablation experiments are also carried out for verifying the effectiveness of proposed modules.

### A. Dataset and Experimental Setup

Due to the lack of data sources, the single look complex (SLC) data provided by OpenSARShip series [47], [48] are the only data that meet our experimental requirements, i.e., containing the polarimetric information. In comparison to OpenSARShip [47], OpenSARShip2.0 [48] provides a more comprehensive and complex ship slicing, which is more aligned with real-world scenarios. Therefore, we here choose it as the experimental dataset, wherein 30 scenes with VV/VH polarization are finally utilized. Similar to the common dataset settings in [7], [18], [19], and [20] on ship classification, a total of 2718 slices are selected from bulk carriers, container ships, and tankers as the ship types for the three-class classification experiment, while a total of 4498 slices involving bulk carriers, cargo, container ships, fishing, general cargo, and tankers are used as the ship types for the six-class classification experiment. The appearance of these types of ships in optical and SAR images is correspondingly presented in Fig. 8. The number of different types of ships is shown in Table I, with 70% of the ships set as the training set and the rest set as the testing set.

TABLE I
Number of Different Types of Ships

| Type | Bulk Carrier | Container Ship | Tanker |
|---|---|---|---|
| Num | 665 | 850 | 1203 |
| Type | Cargo | General Cargo | Fishing |
| Num | 1433 | 241 | 106 |

Our experiments are conducted on the PyTorch framework (version 2.0.1) with the NVIDIA RTX A6000 GPU for acceleration. Meanwhile, we utilize CUDA version 11.7 for GPU computing. The AdamW optimizer [49] is employed with a learning rate of $1e^{-5}$ and a weight decay of 0.05. The cross entropy (CE) loss [50] is chosen as the loss function here. Each experiment is trained for 70 epochs, and the batch size is set to 64. For a more rigorous comparison, we first unify the ship chips by cutting them around the center to 64 × 64, consistent with the ship sizes adopted in the article [21], and then resize them to 224 × 224. Before extracting features, we also normalize the data to avoid significant numerical differences between different features, which simultaneously help the model accelerate the convergence. The compassion methods contain a series of popular networks, that is, Visual Geometry Group 11 (VGG-11) [51], residual network 18 (ResNet-18) [52], dense convolutional network 121 (DenseNet-121) [53], wide residual network 50 (WideResNet-50) [54], EfficientNetV2 [55], SwinTransformerV2 [56], multi-axis vision transformer (MaxViT) [57], ConvNeXt [40], BCNN [17], squeeze-and-excitation laplacian pyramid network with dual-polarization feature fusion (SE-LPN-DPFF) [19], and deep SAR-net (DSN) [21].

### B. Evaluation Criteria

In this article, to evaluate the classification performance of DBDN, the commonly used indicators [18], [19], [20], [34] including accuracy, precision, recall, and F1 score, which are mathematically expressed as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (20)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (21)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (22)$$

$$F1 = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (23)$$

where TP is the number of true positive predictions, TN is the number of true negative predictions, FN is the number of false negative predictions, and FP is the number of false positive predictions.

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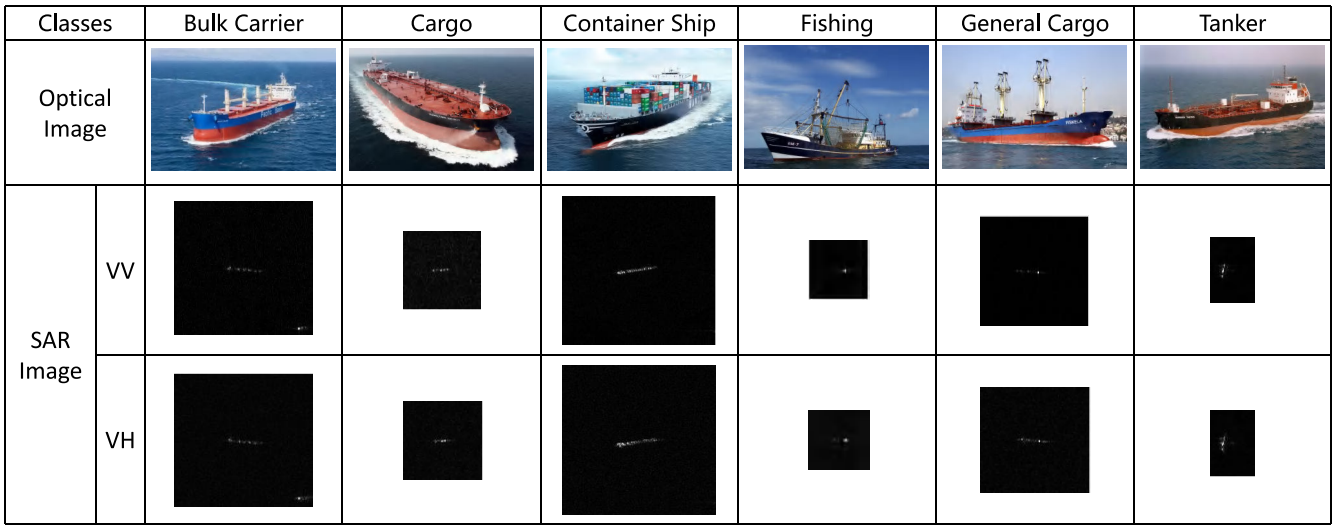



Fig. 8. Optical images versus SAR images. SAR images are from the OpenSARShip dataset, and VV and VH are amplitude maps of SLC products. Note that optical images and SAR images do not correspond to the same ship.

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON OPENSARSHIP2.0. NOTE THAT THE RESULTS OF OUR METHODS ARE MARKED IN BOLD, AND THE BEST RESULTS FROM THE COMPARISON ALGORITHMS ARE UNDERLINED

| Tasks | Three-class Classification | | | | Six-class Classification | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Index(%) / Methods | Accuracy | F1 Score | Precision | Recall | Accuracy | F1 Score | Precision | Recall | Time (s) |
| VGG-11 [51] | 77.70 | 75.48 | 75.92 | 75.22 | 55.70 | 45.79 | 50.74 | 44.75 | 9.31 |
| ResNet-18 [52] | 75.74 | 73.33 | 73.62 | 73.15 | 53.26 | 41.04 | 44.09 | 41.06 | 6.22 |
| DenseNet-121 [53] | 74.75 | 72.88 | 72.91 | 72.85 | 51.70 | 41.43 | 53.35 | 40.59 | 13.21 |
| WideResNet-50 [54] | 74.02 | 71.78 | 71.98 | 71.63 | 48.89 | 40.76 | 46.33 | 39.29 | 12.56 |
| EfficientNetV2 [55] | 76.59 | 74.96 | 75.80 | 74.77 | 55.41 | 45.73 | 54.00 | 44.75 | 20.44 |
| SwinTransformerV2 [56] | 79.29 | 77.81 | 77.83 | 77.80 | 53.78 | 42.17 | 46.95 | 41.32 | 37.64 |
| MaxViT [57] | 77.7 | 75.17 | 76.24 | 74.78 | 53.70 | 42.94 | 51.71 | 43.07 | 27.10 |
| ConvNeXt [40] | 79.29 | 76.84 | 78.33 | 76.23 | 57.41 | 45.58 | 52.29 | 45.54 | 29.26 |
| BCNN [17] | 79.17 | 77.65 | 77.82 | 77.60 | 56.19 | 39.32 | 38.87 | 40.26 | 19.58 |
| SE-LPN-DPFF [19] | 77.94 | 76.45 | 76.86 | 76.31 | 57.75 | 42.26 | 57.97 | 42.34 | 51.78 |
| DSN [21] | 82.88 | 82.18 | 82.38 | 82.58 | 60.19 | 51.99 | 54.32 | 51.37 | 72.20 |
| **DBDN(T*, ours)** | **84.80** | **83.35** | **84.51** | **82.69** | **61.90** | **52.25** | **59.74** | **51.70** | 48.02 |
| **DBDN(P*, ours)** | **86.15** | **85.31** | **85.64** | **85.03** | **63.16** | **53.60** | **58.22** | **52.51** | 48.28 |
| **DBDN(T*&P*, ours)** | **83.95** | **82.83** | **83.19** | **82.59** | **60.42** | **52.45** | **60.36** | **51.81** | 48.17 |
| **DBDN(ours)** | **87.62** | **86.74** | **87.31** | **86.32** | **64.26** | **58.44** | **65.60** | **55.43** | 48.35 |

### C. Experimental Results and Analyses

*1) Quantitative Analyses of DBDN:* All the results of these networks are quantitatively listed in Table II. Note that these values are obtained by averaging ten times.

In detail, none of the first eight networks achieve an accuracy exceeding 80% in the three-class classification experiments. This can be caused by the fact that these networks are primarily designed for tasks involving natural image classification and do not exploit the inherent characteristics of SAR images. Furthermore, while the classification accuracies of BCNN and SE-LPN-DPFF both surpasses that of earlier computer vision classification networks, there are still limitations in their performance. This can be attributed to the fact that the enhancement in classification performance depends not only on the use of additional polarimetric features, but also on the efficacy of the module design in capturing these features. In contrast, DSN achieves a breakthrough in classification accuracy of 82.88%, as it fully utilizes single polarimetric features, i.e., the radar spectrogram. Compared to it, DBDN adopts more dual polarimetric features and texture information as well as performs special module design based on the characteristics of SAR images. Therefore, DBDN shows

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TABLE III
THREE-CLASS CLASSIFICATION CONFUSION MATRIX OF DBDN

| True \ Predicted | Bulk Carrier | Container Ship | Tanker | F1 Score(%) |
|---|---|---|---|---|
| Bulk Carrier | 158 | 18 | 17 | 82.51 |
| Cargo | 13 | 240 | 22 | 88.56 |
| Tanker | 19 | 9 | 320 | 90.52 |
| Precision(%) | 83.16 | 89.89 | 89.14 | $Accuracy =$ |
| Recall(%) | 81.87 | 87.27 | 91.95 | $87.99\%$[1] |

[1] This result is one single experimental realization.

significant improvement in all classification metrics, such as an accuracy increase of 4.74% compared to DSN.

In addition, we also explore the effect of the other three different combination manners of polarimetric and texture feature on the classification performance. The first one is named TF-RGB*, wherein only the LBP operator is replaced with the Canny filter [13] [i.e., the input of DBDN(T*) is "{H, DoD, RS} + {VV amplitude, VH amplitude, $VV_{Canny}$}"]. The second one is called PF-RGB*, which is composed of the degree of linear polarization $m_L$ and the linear polarization ratio $u_L$ derived from the Stokes parameter [24], as well as the mean eigenvalue $\bar{\lambda}$ obtained from the Jones coherence matrix $\boldsymbol{J}$ [58]. At this point, the input of DBDN(P*) is "{$\bar{\lambda}$, $m_L$, $u_L$}+ {VV amplitude, VH amplitude, $LBP_{P,R}$}." The last one is the fusion of PF-RGB* and TF-RGB*, by which the input of DBDN(P*&T*) is "{$\bar{\lambda}$, $m_L$, $u_L$} + {VV amplitude, VH amplitude, $VV_{Canny}$}."

From the experimental results in Table II, it is easily found that these new combinations are not as effective as the original combination we selected. This is because the polarimetric characteristics we chose can reflect the characteristics of ships from different aspects. Specifically, $H$ provides insights into the variability and randomness of backscattering, DoD capitalizes on material-based differences in depolarization, and RS offers robust structural discrimination and environmental resilience. While Canny filtering can highlight the edge features of images, it is not as effective as the LBP operator in recognizing complex texture patterns. Even so, these new feature combinations still outperform the baselines, demonstrating the effectiveness of introducing polarimetric and texture features, as well as the superiority of the DBDN architecture we designed. These results also suggest the need to carefully consider the overall effectiveness and task-specific suitability of polarimetric and texture features when making selections.

The last column of Table II records the average time for each algorithm to run one epoch on the three-class classification experiment. From Table II, it can be summarized that as the depth and width of network increase, the running time also increases. Although DSN has achieved performance breakthroughs compared to other algorithms, its time consumption is the most. Meanwhile, it is found that DBDN has a slightly longer run time than computer vision algorithms due to the extraction of more features. Even so, it is still faster than the recently proposed methods SE-LPN-DPFF and DSN. Thus, from the viewpoint of balancing performance and time consumption, DBDN is still one satisfactory network.

Thus, the effectiveness of DBDN in ship classification is verified. The same conclusion can be attained via further analyzing the quantitative results of six-class classification. Note that, for these values, they are all lower than three-class classification. This is because the number of fishing and general cargo is very small compared to the other types of vessels, i.e., the category imbalance problem has a significant impact on the classification results of six-class classification.

*2) Classification Confusion Matrix of DBDN:* Tables III and IV, respectively, display the three-class and six-class confusion matrices of DBDN. From Table III, it is evident that DBDN can accurately classify the majority of ships, as the diagonal values are higher than the other values in the same row. In Table IV, despite the presence of highly imbalanced samples, DBDN still achieves an F1 score of over 60% except for general cargo. This could be attributed to the fact that general cargo shares certain similarities with other categories in terms of hull shape, cabin layout, and other aspects. Consequently, in SAR images with limited resolution, it becomes easier for general cargo to be mistaken for other ship types.

*3) Performance Analyses of PF-RGB:* From previous experiments, it is evident that DBDN which utilizes polarimetric features greatly enhances the accuracy of ship classification. As most SAR ship classification networks only employ the VV amplitude map as the input data, it is imperative to explore the impacts of PF-RGB and single VV amplitude map on classification results from the perspective of input.

Fig. 9 illustrates the comparison between PF-RGB and VV amplitude map on different baselines. Clearly, for all baselines, when PF-RGB is adopted as the input, their quantitative results are greater in both three-class and six-class classification tasks. Taking VGG-11 as an example, PF-RGB achieves an accuracy improvement of 2.45% and 2.23% in the three-class and six-class experiments, respectively. The reason is that PF-RGB comprises of three different polarimetric features that can offer more diverse structural characteristics and multichannel spatial information than the single VV amplitude map. Hence, employing PF-RGB as the input can effectively improve the classification performance of networks.

*4) Performance Analyses of TF-RGB:* As mentioned earlier, DBDN also utilizes TF-RGB as the input. Therefore, we here also explore the parameter ($P$, $R$) effect of $LBP_{P,R}$ in ship

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TABLE IV
SIX-CLASS CLASSIFICATION CONFUSION MATRIX OF DBDN

| True \ Predicted | Bulk Carrier | Cargo | Container Ship | Fishing | General Cargo | Tanker | F1 Score(%) |
|---|---|---|---|---|---|---|---|
| Bulk Carrier | 120 | 48 | 16 | 0 | 2 | 10 | 65.93 |
| Cargo | 20 | 302 | 10 | 0 | 23 | 80 | 63.98 |
| Container Ship | 12 | 12 | 210 | 0 | 8 | 11 | 80.31 |
| Fishing | 0 | 14 | 1 | 15 | 1 | 4 | 60.00 |
| General Cargo | 9 | 16 | 12 | 0 | 17 | 15 | 26.15 |
| Tanker | 7 | 117 | 21 | 0 | 10 | 207 | 60.09 |
| Precision(%) | 71.43 | 59.33 | 77.78 | 100.00 | 27.87 | 63.30 | $Accuracy =$ |
| Recall(%) | 61.22 | 69.43 | 83.00 | 42.86 | 24.64 | 57.18 | $64.52\%$[1] |

[1] This result is one single experimental realization.

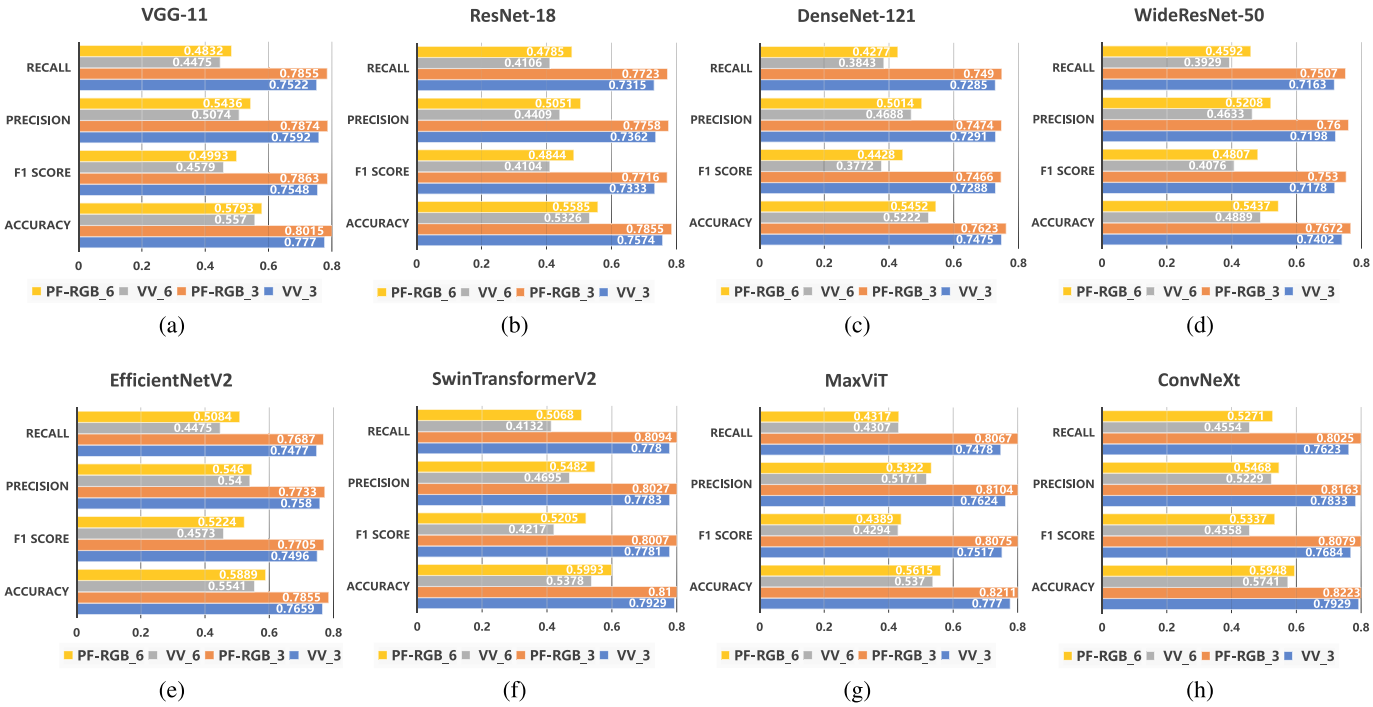



Fig. 9. Comparative experiments of PF-RGB and VV amplitude map as inputs. In the legend, _3 and _6 represent three-class and six-class classification experiments, respectively. (a) VGG-11. (b) ResNet-18. (c) DenseNet-121. (d) WideResNet-50. (e) EfficientNetV2. (f) SwinTransformerV2. (g) MaxViT. (h) ConvNeXt.

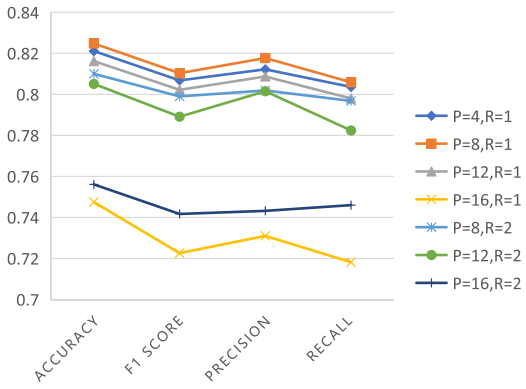



Fig. 10. Impact of different ($P$, $R$) parameters of LBP on the three-class classification task.

classification, whose results are correspondingly presented in Fig. 10. Obviously, by increasing the value of $P$ to 8 and reducing the value of $R$ to 1, the $\text{LBP}_{P,R}$ operator is able to capture more intricate texture information and spatial distributions of textures. So, the best combination (8, 1) is exploited in our experiments.

*5) Performance Analyses of Blocks:* Hereinafter, three experiments are, respectively, conducted to evaluate the effectiveness of the blocks (i.e., CNABlock, EIRBlock, and CSAM) designed for DBDN. Similarly, only the results of three-class classification are shown here to save space.

Table V shows the experimental results before/after replacing CNABlock with CNBlock in DBDN. Clearly, the introduction of attention mechanism in CNBlock has a significant improvement on the classification result. Table VI presents the quantitative results between IRBlock and EIRBlock, from which the effectiveness of EIRBlock can be easily verified. Table VII displays the results of feature fusion stage using SE, CBAM, and our proposed CSAM. It is appearance that the use of CBAM results in an inferior classification performance compared to SE. This can be owing to that CBAM employs

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TABLE V
COMPARATIVE EXPERIMENTS OF THREE-CLASS BEFORE/AFTER REPLACING CNABLOCK WITH CNBLOCK IN DBDN

| Type of Block | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|
| CNBlock | 84.93 | 83.55 | 84.71 | 82.87 |
| CNABlock | **87.62** | **86.74** | **87.31** | **86.32** |

TABLE VI
COMPARATIVE EXPERIMENTS OF IRBLOCK AND EIRBLOCK: THREE-CLASS EXPERIMENTS USING DBDN

| Type of Block | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|
| IRBlock [44] | 86.40 | 85.32 | 85.99 | 84.86 |
| EIRBlock | **87.62** | **86.74** | **87.31** | **86.32** |

TABLE VII
COMPARATIVE EXPERIMENTS OF SE, CBAM, AND CSAM: THREE-CLASS EXPERIMENTS USING DBDN

| Type of Block | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|
| SE [45] | 86.64 | 85.66 | 87.20 | 84.75 |
| CBAM [46] | 85.66 | 84.75 | 85.41 | 84.26 |
| CSAM(ous) | **87.62** | **86.74** | **87.31** | **86.32** |

both adaptive max pooling and adaptive average pooling, where adaptive max pooling is not suitable for extracting ship features. Even so, the SE module does not perform as well as CSAM. This is mainly due to the absence of spatial attention, which hinders the model's ability to focus on the spatial information.

Overall, these three experiments provide evidences that our designed blocks are indispensable for enhancing the classification performance of DBDN.

### D. Ablation Study

*1) Ablation Study on Polarimetric Features:* We conduct an ablation study on the three used polarimetric features, i.e., DoD, $H$, and RS. Take the three-class classification as an example; the results are shown in Table VIII. Compared to using each feature individually, combining two features together can achieve better performance. Furthermore, in comparison with the fusion of two features, fusing all these three features enables us to obtain the best classification results. So, this directly implies that the utilization of polarimetric information (i.e., PF-RGB) is really helpful for ship classification.

*2) Ablation Study on Texture Features:* The ablation experiments' results of three-class experiment of texture features are shown in Table IX. There, it is apparent that extra incorporating $\text{LBP}_{P,R}$ features indeed leads to greater values compared to the single combination of VV and VH amplitude maps. So, this verifies that the textural information brought by $\text{LBP}_{P,R}$ is really effective in ship classification.

*3) Ablation Study on Dual Branch Inputs:* The ablation experiments' results of three-class experiment of dual branch features are shown in Table X. It can be seen that compared to using only PF-RGB or TF-RGB as input, using dual branch RGB images as input further improves the performance of ship classification. Thus, the strategy that combining polarimetric information and texture information together for ship classification is effective.

TABLE VIII
ABLATION STUDY ON POLARIMETRIC FEATURES: THREE-CLASS EXPERIMENTS USING CONVNEXT [40]

| DoD | H | RS | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|---|---|
| ✓ | - | - | 79.29 | 77.39 | 78.01 | 77.01 |
| - | ✓ | - | 78.80 | 76.74 | 77.71 | 76.26 |
| - | - | ✓ | 79.04 | 77.96 | 78.10 | 77.87 |
| ✓ | ✓ | - | 81.62 | 80.21 | 80.31 | 80.29 |
| ✓ | - | ✓ | 80.27 | 79.27 | 79.03 | 79.85 |
| - | ✓ | ✓ | 81.62 | 80.43 | 80.80 | 80.26 |
| ✓ | ✓ | ✓ | **82.72** | **81.55** | **81.77** | **81.38** |

TABLE IX
ABLATION STUDY ON TEXTURE FEATURES: THREE-CLASS EXPERIMENTS USING CONVNEXT [40]

| VV | VH | LBP | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|---|---|
| ✓ | - | - | 79.29 | 76.84 | 78.33 | 76.23 |
| - | ✓ | - | 77.94 | 75.52 | 75.16 | 76.17 |
| ✓ | ✓ | - | 81.13 | 79.42 | 81.59 | 78.39 |
| ✓ | ✓ | ✓ | **82.48** | **81.03** | **81.77** | **80.59** |

TABLE X
ABLATION STUDY ON DUAL BRANCH FEATURES INPUTS: THREE-CLASS EXPERIMENTS, WHERE SINGLE BRANCH FEATURES USE CONVNEXT [40], AND DUAL BRANCH FEATURES USE BASE NETWORK

| PF | TF | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|---|
| ✓ | - | 82.72 | 81.55 | 81.77 | 81.38 |
| - | ✓ | 82.48 | 80.91 | 81.28 | 80.62 |
| ✓ | ✓ | **85.42** | **84.15** | **84.85** | **83.71** |

*4) Ablation Study on Network Architecture:* In Tables XI and XII, we conduct ablation experiments on the network architecture. The network "Base" in both tables indicates the basic network, i.e., the ICM and FEM without CNABlock. To a certain degree, the architecture of DBDN is also gradually derived from "Base."

In Table XI, the ablation experiment is first performed on the impact of pretraining. Clearly, compared to the fine-tuning with pretrained weights, the classification results obtained by initializing network weights with Kaiming initialization from scratch are limited, with only 72.06% classification accuracy. So, in terms of small sample data, using the transfer learning technology to utilize the generalized features obtained from large datasets is necessary. Hence, it is reasonable for us to take advantage of this technology in DBDN. Through comparing "Base" and "Base + CNABlock," one can easily see that CNABlock holds the ability to further improve the performance of "Base." For "Base + CNABlock + EIRBlock," the module EIRBlock leads to an improvement in accuracy from 86.15% to 87.13%. After adding the module CSAM to "Base + CNABlock + EIRBlock," we can find that DBDN achieves the best classification accuracy of 87.62%. Directly, the effectiveness of proposed fusion strategy is verified. Since the same conclusion can be attained from Table XII, the

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TABLE XI
ABLATION STUDY OF DBDN IN THREE-CLASS EXPERIMENTS. NOTE THAT THE BEST RESULTS ARE MARKED IN BOLD

| Network Architecture | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|
| Base | 85.42 | 84.15 | 84.85 | 83.71 |
| Base(Training from scratch with Kaiming initialisation) | 72.06 | 68.48 | 71.74 | 67.87 |
| Base + CNABlock | 86.33 | 85.36 | 85.51 | 85.30 |
| Base + CNABlock + EIRBlock | 87.13 | 86.15 | 86.37 | 85.95 |
| Base + CNABlock +EIRBlock + CSAM (DBDN) | **87.62** | **86.74** | **87.31** | **86.32** |

TABLE XII
ABLATION STUDY OF DBDN IN SIX-CLASS EXPERIMENTS. NOTE THAT THE BEST RESULTS ARE MARKED IN BOLD

| Network Architecture | Accuracy(%) | F1 Score(%) | Precision(%) | Recall(%) |
|---|---|---|---|---|
| Base | 61.78 | 55.56 | 58.81 | 54.10 |
| Base(Training from scratch with Kaiming initialisation) | 48.70 | 34.78 | 35.47 | 36.38 |
| Base+ CNABlock | 62.59 | 55.40 | 64.72 | 52.51 |
| Base + CNABlock + EIRBlock | 63.41 | 55.69 | 61.23 | 53.54 |
| Base + CNABlock +EIRBlock + CSAM (DBDN) | **64.26** | **58.44** | **65.60** | **55.43** |

detailed analyses of six-class experiment are not given here for saving space.

## V. CONCLUSION

In this article, a novel DL network named DBDN was proposed for ship classification in dual-polarized SAR image. It included an ICM for capturing intricate ship information from both polarimetric and texture perspectives, an FEM for addressing the issue of limited data and extracting ship characteristics via transfer learning and attention mechanisms, and an FFCM for further extracting and integrating deeper features to classify ships. Experimental results from the OpenSARShip2.0 dataset demonstrate DBDN's effectiveness. In particular, compared to the recently proposed SOTA algorithm DSN, DBDN holds a much better classification performance. Taking the metric Accuracy for instance, DBDN further improved it by 4.74% and 4.07% in the three-class and six-class experiments, respectively.

Although DBDN exhibits a satisfactory ship classification performance, future researches are still needed to concentrate on the application of more different combinations of polarimetric features and texture features to ship classification.

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